

Advanced Linux Troubleshooting Commands – Full Detailed Guide

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NETWORK TROUBLESHOOTING

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1. netstat -tulnpa

- netstat: Displays network connections and ports.
- -t: TCP ports
- -u: UDP ports
- -l: Listening ports
- -n: Show numeric IPs instead of DNS names
- -p: Show process using the port
- -a: Show all connections

2. ss -tulnw

- ss: Modern replacement for netstat, faster.
- -t: TCP
- -u: UDP
- -l: Listening
- -n: Numeric output
- -w: Raw sockets

3. tcpdump -i eth0

- Captures network packets on a specific interface.
- Use filters like:

tcpdump -i eth0 port 80

tcpdump -i eth0 host 10.0.0.5

4. dig

- DNS lookup tool.

- dig google.com ANY
- dig @8.8.8.8 google.com

5. nslookup

- Basic DNS lookup.

6. curl -v

- Tests web server responses.
- -v: Verbose output

7. traceroute

- Shows the path packets take to reach a host.

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SYSTEM & PROCESS TROUBLESHOOTING

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8. top

- Shows CPU, memory, processes in real-time.

9. htop

- Interactive top with colors and filtering.

10. ps auxf

- Shows running processes in a tree format.

11. pstree -p

- Shows a tree view of processes with PIDs.

12. kill -9

- Force kills a process.

13. renice -n -p

- Changes process priority:

- Negative = higher priority
- Positive = lower priority

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MEMORY TROUBLESHOOTING

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14. free -m

- Displays memory usage in MB.

15. vmstat 1 5

- Reports memory, CPU, IO, process statistics.

16. sar -r 1 5

- Memory stats over time (sysstat package).

17. cat /proc/meminfo

- Detailed memory statistics.

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DISK & FILESYSTEM TROUBLESHOOTING

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18. df -Th

- Shows filesystem type and usage.

19. du -sh /path/*

- Shows size of directories.

20. lsblk -f

- Shows block devices, filesystems, UUIDs.

21. mount | column -t

- Shows mounted filesystems.

22. blkid

- Shows block device attributes.

23. smartctl -a /dev/sda

- Checks disk health (smartmontools).

24. iostat -x 1 5

- Shows disk IO stats (sysstat package).

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LOGS & ERROR DIAGNOSTICS

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25. journalctl -xe

- Shows system logs with errors.

26. tail -f /var/log/syslog

- Live logs for debugging.

27. dmesg | tail -50

- Shows recent kernel messages.

28. grep -i error /var/log/*

- Searches error messages in logs.

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SERVICE TROUBLESHOOTING (systemd)

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29. systemctl status

- Shows service status & logs.

30. systemctl restart

- Restarts a service.

31. systemctl enable

- Enables service at boot.

32. systemctl list-units --failed

- Shows failed services.

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NETWORK CONFIGURATION & CONNECTIVITY

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33. ip addr

- Shows network interfaces.

34. ip route

- Shows routing table.

35. ip neigh

- Shows ARP table.

36. ifconfig -a

- Shows interface configuration (legacy).

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USER & PERMISSION TROUBLESHOOTING

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37. id

- Shows user/group IDs.

38. whoami

- Shows current user.

39. getfacl

- Shows file ACL permissions.

40. setfacl -m u:user:rwx

- Modifies ACL permissions.

41. chattr +i

- Makes file immutable (cannot modify/delete).

42. lsattr

- Shows file attributes.

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PACKAGE & SYSTEM HEALTH

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43. rpm -qa | grep

- Checks installed packages (CentOS/RHEL)

44. dpkg -l | grep

- Checks installed packages (Ubuntu)

45. systemctl list-timers

- Shows scheduled system tasks.

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ADVANCED TROUBLESHOOTING

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46. strace -p

- Shows system calls made by a process.

47. lsof

- Lists open files and sockets.

48. lsof -i :80

- Shows process using port 80.

49. nmap

- Scans ports on a system.

50. sysctl -a

- Shows kernel parameters.

51. uptime

- Shows system load.

52. lastb

- Shows failed login attempts.

53. fail2ban-client status

- Shows blocked IPs (if Fail2Ban installed).

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CONTAINER TROUBLESHOOTING (BONUS)

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54. docker ps -a

- Shows all containers.

55. docker logs

- Shows container logs.

56. docker inspect

- Shows container configuration.

57. docker exec -it bash

- Enters a container shell.

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SUMMARY

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This document includes both basic and advanced Linux troubleshooting commands, detailed explanations, and real-world usage. It is suitable for DevOps engineers, SysAdmins, Support teams, and Linux learners.